

Problem 1. Consider the equation

$$\lfloor x \rfloor (x^2 + 1) = x^3,$$

where $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

- Prove that the equation has infinitely many positive real solutions.
- Show that none of the positive real solutions of this equation is rational.

Problem 2. Let n be a positive integer. Assume that n numbers are to be chosen from the table

$$\begin{array}{cccc} 0 & 1 & \cdots & n-1 \\ n & n+1 & \cdots & 2n-1 \\ \vdots & \vdots & \ddots & \vdots \\ (n-1)n & (n-1)n+1 & \cdots & n^2-1 \end{array}$$

with no two of them from the same row or the same column. Find the maximum value of the product of these n numbers.

Problem 3. There are 2018 lamps arranged on a table. Two persons play the following game. In each move the player flips the switch of one lamp, but he must never get back an arrangement of the lit lamps that had happened before. A player who cannot move loses. Which player has a winning strategy?

Problem 4. Let T denote the 15-element set $\{10a + b : a, b \in \mathbb{Z}, 1 \leq a < b \leq 6\}$. Let S be a subset of T in which all six digits $1, 2, \dots, 6$ appear and in which no three elements together use all these six digits. Determine the largest possible size of S .

Problem 5. Let AB and CD be two diameters of the circle Γ . For an arbitrary point P on Γ , let R and S be the feet of the perpendiculars from P to AB and CD , respectively. Show that the length of RS is independent of the choice of P .

Problem 6. In the non-isosceles triangle ABC an altitude from A meets side BC in D . Let M be the midpoint of BC and let N be the reflection of M in D . The circumcircle of triangle AMN intersects the side AB in $P \neq A$ and the side AC in $Q \neq A$. Prove that AN , BQ and CP are concurrent.

Problem 7. For any integer $n \geq 2$, we define A_n to be the number of positive integers m with the following property: the distance from n to the nearest multiple of m is equal to the distance from n^3 to the nearest multiple of m . Find all integers $n \geq 2$ for which A_n is odd. (Note: The distance between two integers a and b is defined as $|a - b|$.)

Problem 8. An integer $n \geq 1$ is called *balanced* if it has an even number of distinct prime divisors. Prove that there exist infinitely many positive integers n such that there are exactly two balanced numbers among $n, n+1, n+2$ and $n+3$.